HIKET — the storyline

a soil that carries its history: the trilemma of the balanced start, tested in Finland

Version of October 13, 2026. Numbers restated on 2026-10-13 on the recalibrated pair with the 1961–1984 pre-run climate: production `20261012_1042*`; balanced start `20261012_12*-13*`.

How to read this. A basis for the story discussion, not a draft. Each section opens with a strip of the whole paper, its own place highlighted, and then its internal arc. In the diagrams: blue = the section's payload; violet = a thread of the trilemma; amber fill = a caveat that travels with a claim. Figures sit where they are used: captions say *main text*, **appendix** or **not in your list: keep?**. Green boxes are the questions still open.

The decisions this storyline is built on (Lorenzo, 2026-10-12).

- **General first, Finland as the test.** The claim opens on the general problem — soil carbon carries a long history, a balanced start assumes it away, and that decides the forecast — and Finland is the field where the general statement is tested with measurements.
- **The trilemma is the Introduction's connecting object.** Three sources of information — the soil series, the kinetics, the inputs — must agree, and a balanced start satisfies two at most. Published kinetics are a source of information like the others. The threads of the paper meet there.
- **The trilemma rests on the runs without the transient start**, even though those cannot appear in the Introduction: it is introduced there as a trilemma, and contextualised as soon as the Results arrive — **F16, the trilemma visualised** (R1). It comes back once more when the ridge appears (F18, R2), the same triangle in parameter space. The schematic need not be a main figure.
- **The balanced start**: F16 in R1 as the trilemma visualised; its rates (F17) and forecast (F19) in the appendix. **S16 is a main figure**.
- **R2 is the main arc: the resolution.** R1 opens and defines the problem, tested the hard way; from R1 to R2 runs the justification of the central choice — relaxing the steady-state assumption — which moves the models across the ridge. Everything after (forecast, plots) is built on that choice. F3 and S16 belong to R2.
- **The uncertainty is a nested side arm, replicated at the end of R1 and R2.** R1 (balanced start) has σ_{input} only, absorbing the uncertainty of the input estimate (it connects to the data discussion and S18). R2 (estimated start) has σ_{input} and σ_{init}: σ_{init} becomes needed — its need was hidden in R1 by forcing the steady-state assumption, which fixes it at 1 without testing it. R2 is not more uncertain; it makes explicit what R1 assumed. σ_{init} is the development of R2 from the general idea to its implementation.
- **The forecast is an important side arm**, close to the other parts in weight but not quite, and not in the title.
- **The plots are a separate topic**, after the model fit and its R^2, tied to the trilemma, and leading into a Discussion of how the models could do better.

The story on one page

0.0.0.1 The general claim. Soil carbon carries a long history. Its slowest pools adjust to a change in their input over centuries, much longer than the few decades a unit of carbon spends in the soil

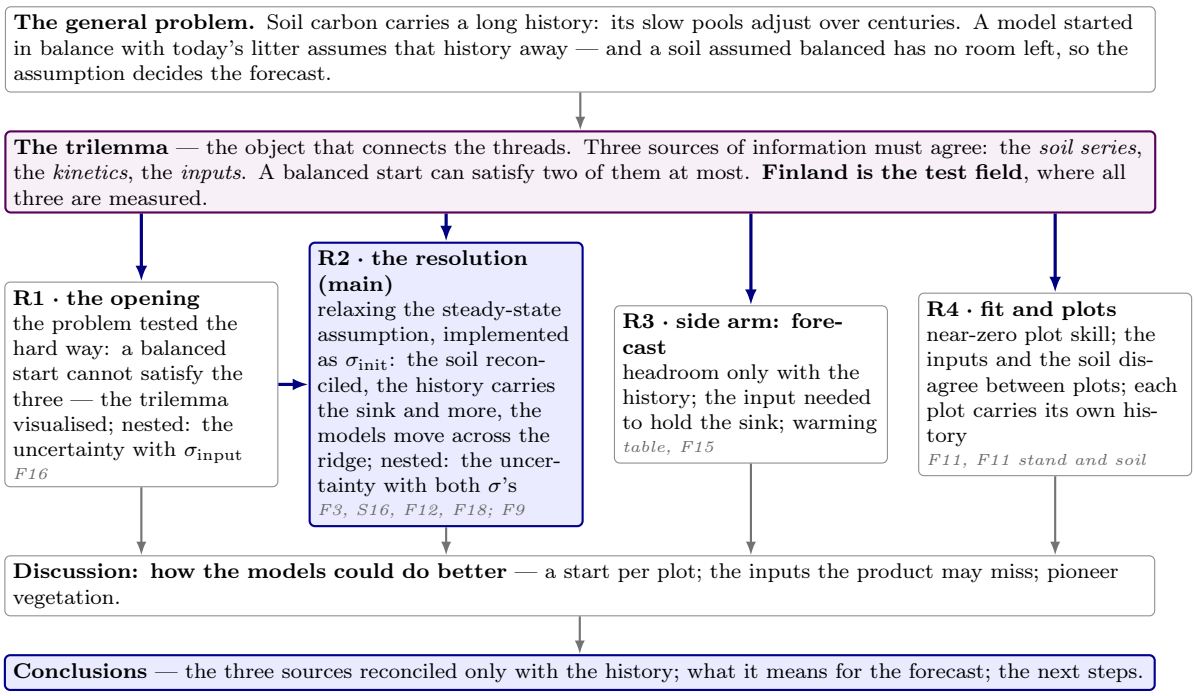
on average, so the soil of today still reflects the land use and the inputs of long before any model starts. The convention is to start soil models in balance with their present litter, which assumes that history away. **And the assumption decides the forecast**: a soil assumed balanced has no room left to gain, so its projected sink depends only on future changes in litter and climate, while a soil still catching up with its past keeps a sink of its own.

0.0.0.2 The connecting object: a trilemma. Any model of a soil's trajectory has to reconcile three sources of information: the measured *soil series*, the *kinetics* (decomposition rates and climate response, themselves the product of earlier calibrations against measurements), and the measured *inputs*. A model started in balance can satisfy two of them at most. The threads of the paper meet there: the trade-off between input and turnover, the missing input, the history of each plot, and the forecast.

0.0.0.3 The test field. Finland is where all three are measured on the same ground: a landscape reshaped by a century of land use and regrowth, three soil campaigns on the same plots over four decades, a plot-level litter record checked against the national inventory, and published kinetics for the inventory's own model family. On the same models and the same data, a balanced start cannot reproduce the measured accumulation, a start estimated from the measurements brackets it, and the modelled sink of 1985–2024 is then the soil still catching up with its own history: the inherited part exceeds the whole sink, because the warming takes back more than the litter rise adds. Plot by plot, the same history is what the models miss.

0.0.0.4 The consequences. Relaxing the assumption changes what the models say about the future of the Finnish forest soil sink: with the history the soil keeps room to gain, about a tenth more litter input would hold the sink of the past for twenty years, and warming takes part of it back — a direct input to the debate on the national greenhouse gas balance, which a soil assumed balanced cannot inform.

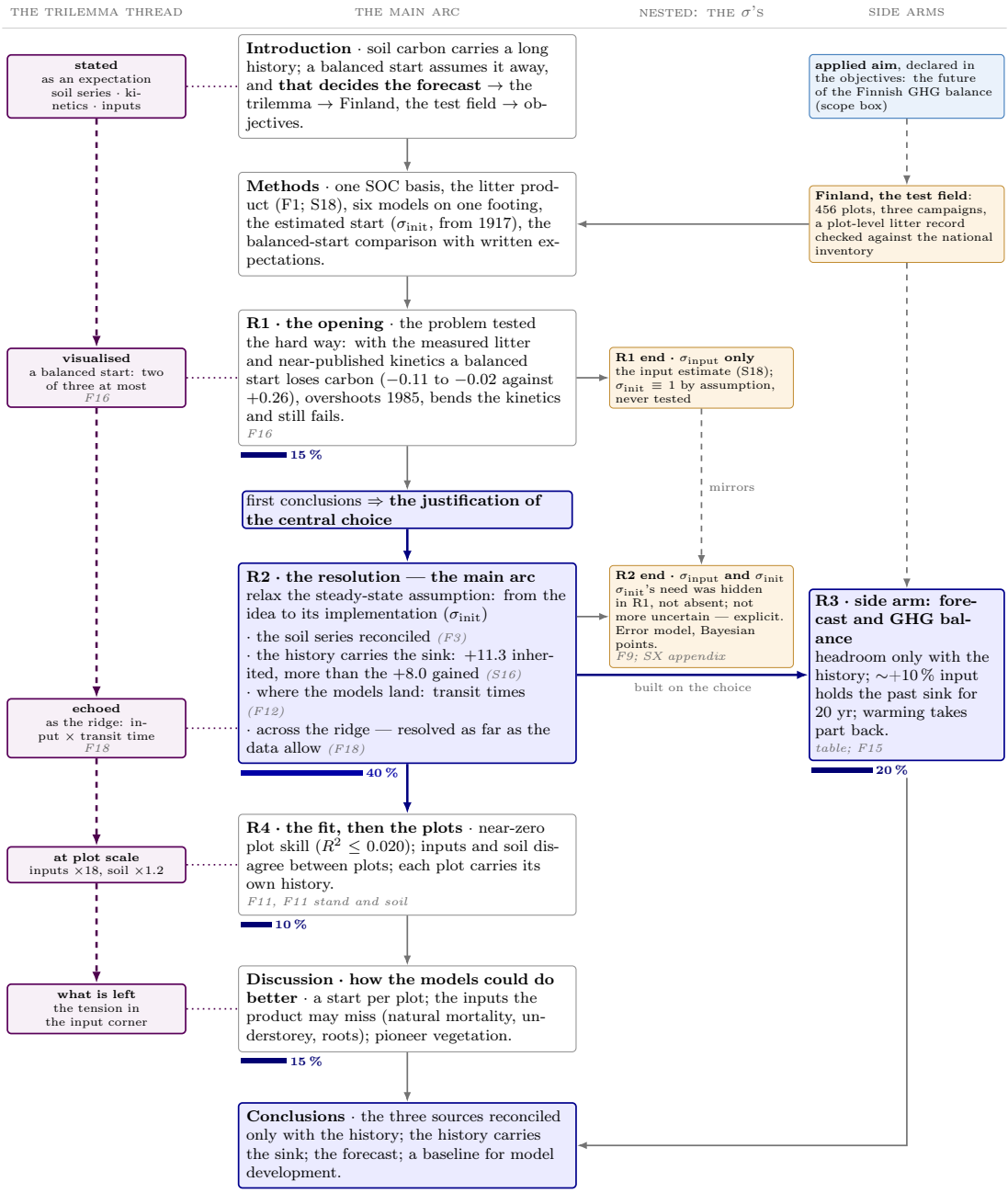
0.0.0.5 The arc of the whole paper.



The weight budget. Proposed: **R1** the opening ∼15%; **R2** the resolution ∼40% — the main arc; **R3** the forecast side arm ∼20%; **R4** fit and plots ∼10%, leading into a **Discussion** with real space (∼15%).

The storyline schema

The whole paper on one page, in four lanes. **The main arc** runs down the middle: the Introduction states the general problem, R1 tests it the hard way, and R2 resolves it by relaxing the steady-state assumption; R4 and the Discussion follow from it. **The trilemma thread** (left, violet) is stated in the Introduction and comes back at each step where it is visible: in R1 as the failure of the balanced start (F16), in R2 as the ridge (F18), in R4 at plot scale, and in the Discussion as what is left. **The nested σ side arm** closes R1 and R2 and mirrors itself. **The forecast side arm** (R3) is built on the central choice and is the place where the applied aim of the objectives is answered. Bars give the proposed weight of each section.

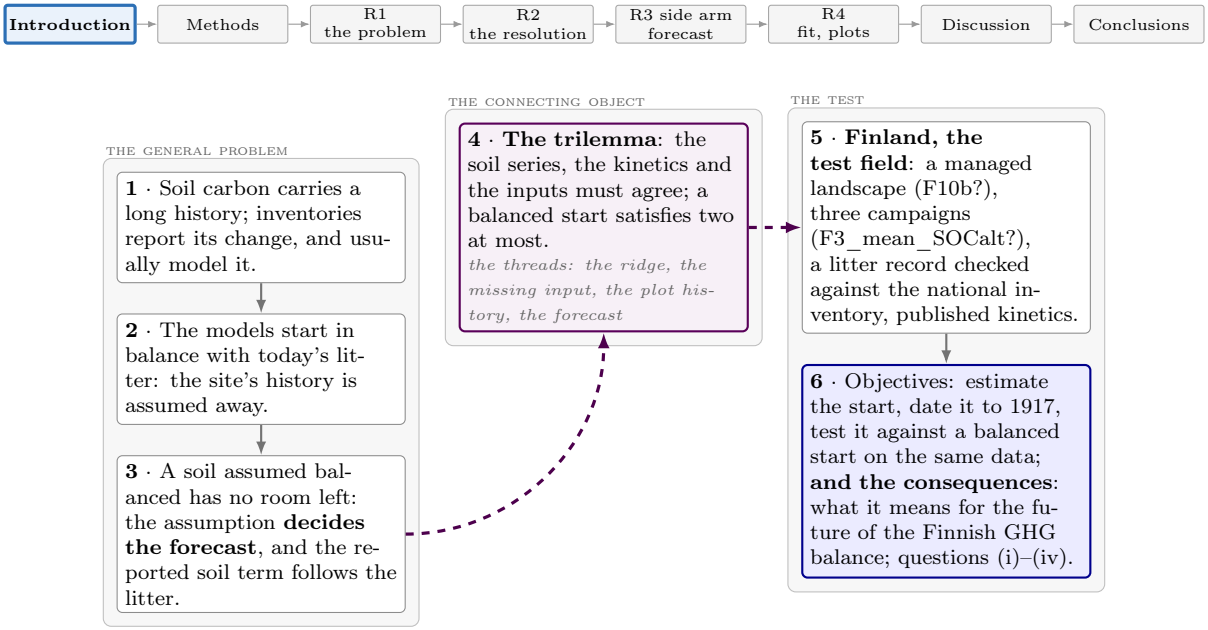


The figure plan

where	figure	role	status
Introduction	F10b_forest_history	the landscape was not in balance	**not in your list**
Introduction	F3_mean_SOCalt	the measured soil, model-free	**not in your list**
Methods	F1_inputs	the litter series, one corner of the trilemma	main
Methods	*S18*	litter: plot product vs national inventory, same scope	appendix
Methods	F2_baseline	one homogenised SOC basis	main
Methods	F4_initialization	how the estimated start is built	main
R1	F16_eqinit_trajectories	the balanced start against the campaigns — the trilemma visualised	main
R1	*F17*	the balanced start, rates by interval	appendix
R2	F3_mean_soc	the estimated start against the campaigns	main
R2	S16_attribution_ensemble	how much of the sink is the history	main
R2	*S15*	the same, per model	appendix
R2	F9_auxiliary_sigmas	the two auxiliary parameters (nested part, end of R2)	main
R2	*SX*	σ_{input} against independent estimates (nested part)	appendix
R2	F12_mrt_yasso	transit time, calibrated and published	main
R2	F18_eqinit_tradeoff	the ridge, the two starts on it — the trilemma's echo	main
R2	*F14*	the ridge on the likelihood surface	appendix
R3	*F19*	the forecast from the two starts (prose)	appendix
R3	*table*	the input that would hold the sink	main
R3	F15_forward_arms	the warming side arm	main
R4	*F6*	predicted vs measured on held-out plots (R^2 in the text)	appendix
R4	F11_rf_heatmap	what the residual follows	main
R4	F11_stand_and_soil	the stand of 1985 and the soil	main
Discussion	*F11b*	time since the cut, behind a start per plot	appendix

Still open: F10b and F3_mean_SOCalt, the two Introduction figures of the current draft. The test field needs the soil series on the table before the objectives, so F3_mean_SOCalt is probably still needed; F10b carries the landscape history.

1 Introduction



1.0.0.1 1. Soil carbon carries a long history, and its change is usually modelled.

Inventories report the annual change in the soil carbon stock, not the stock. A few countries measure it (Sweden, Germany); most model it, Finland with Yasso07. The slow pools adjust over centuries, while the mean transit time is decades (Sierra et al. 2018): a soil keeps the memory of its past inputs for far longer than its turnover suggests.

1.0.0.2 2. The convention assumes that history away.

Models are started in balance with a constant litter and climate — the Finnish inventory, the nationwide Yasso07 evaluation, RothC, the land models — and the inventory justifies it in one sentence ("approximately ten years of simulation since spin-up is enough to cancel out the effect of the spin-up level"). A balanced start responds to every change after its start date, but it cannot carry a deficit that already existed on that date: the site's history is replaced by an equilibrium with its present litter.

1.0.0.3 3. The assumption decides the forecast.

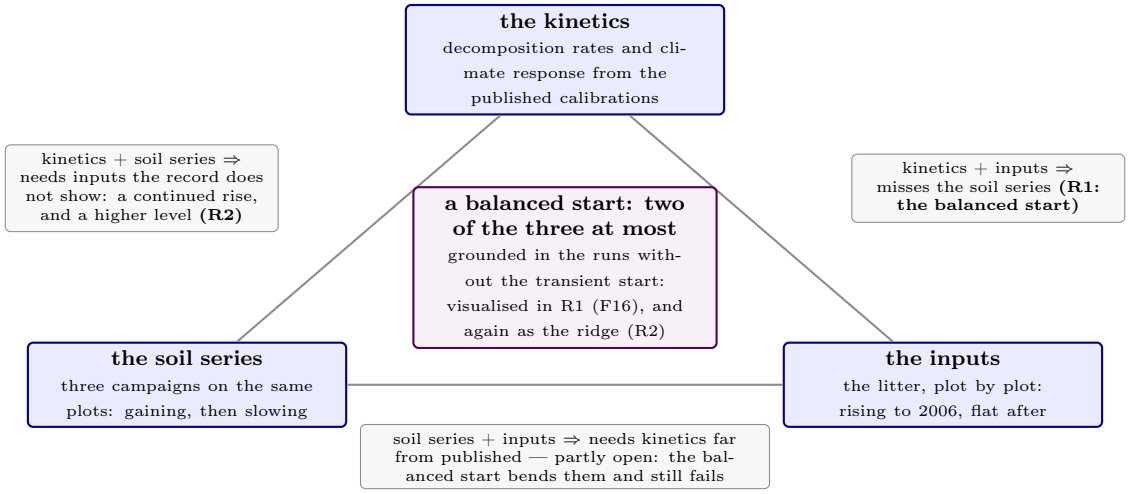
A soil assumed balanced has no room left to gain: its projected sink comes only from future changes in litter and climate. The reported Finnish soil term shows what that means in practice: a sink for most of the reporting period, a source from 2021 once the litter input was recalculated — in a model started in balance, the soil term follows the litter estimate.

> **Tone.** Describe the method, never judge the inventory (Aleksi leads it and is a coauthor). The safe sentence: *the reported soil term is, in both directions, the response of a modelled soil to a litter input that departs from the level the soil was started in balance with.*

1.0.0.4 4. The trilemma — where the threads meet.

A model of a soil's trajectory has three sources of information to reconcile: the *soil series* (repeated measurements of the stock), the *kinetics* (decomposition rates and climate response, calibrated against measurements in earlier work), and the *inputs* (the litter, measured or derived plot by plot). In a soil in balance, the stock is the input times the transit time, so the three are tied together: given the inputs and the kinetics, the stock

and its change follow, and a soil that is gaining carbon while its litter has stopped rising cannot be in balance with that litter. A balanced start can therefore satisfy two of the three at most. The paragraph should be short; the triangle carries the argument, and the threads it connects are named once here and come back later: the trade-off between input and turnover (R2), the missing input (R2, Discussion), the history of each plot (R4), and the forecast (R3). **The trilemma rests on the runs without the transient start**; they cannot appear in the Introduction, so the trilemma is stated there as an expectation and contextualised as soon as the Results begin (R1, F16).



1.0.0.5 5. Finland, the test field. Where the general statement becomes measurable. A managed landscape far from balance: slash-and-burn, grazing and high-grading, the war decades, then a century of regrowth (+84% growing stock) (Liski et al. 2006; Peltoniemi et al. 2004, 2007 said this in the group's own words). Three soil campaigns on the same plots: 64.5 → 71.5 → 73.6 tC ha^{-1}, gaining and then slowing. A litter record for every plot (Tupek et al.), rising by about 45% to 2006 and flat after, which on the scope it shares with the national inventory agrees with it where most plots lie. And published kinetics for the inventory's own model family. All three corners of the trilemma are measured on the same ground. ▶ F10b_forest_history

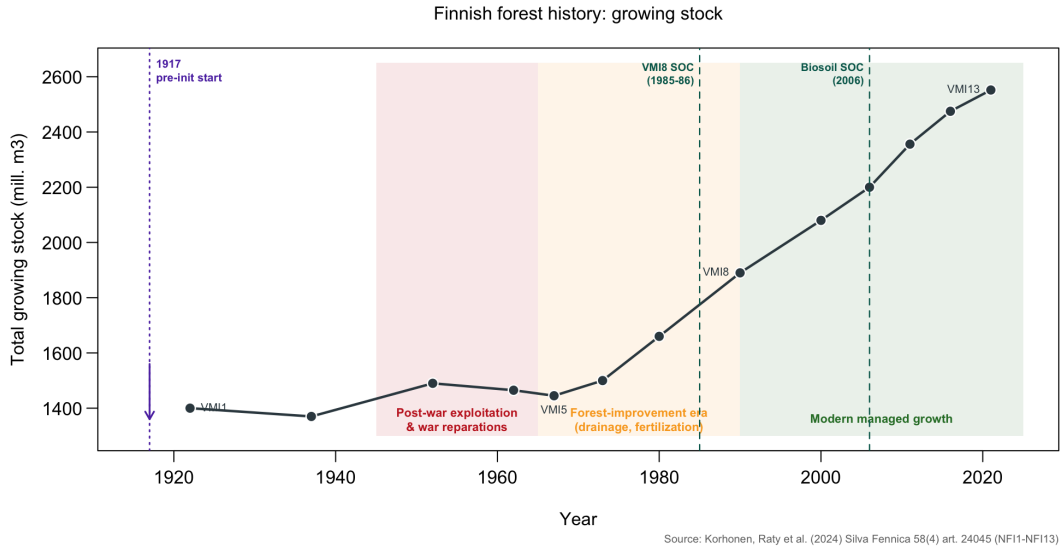


Figure 1: F10b_forest_history — **not in your list: keep?**

▶ F3_mean_SOCalt

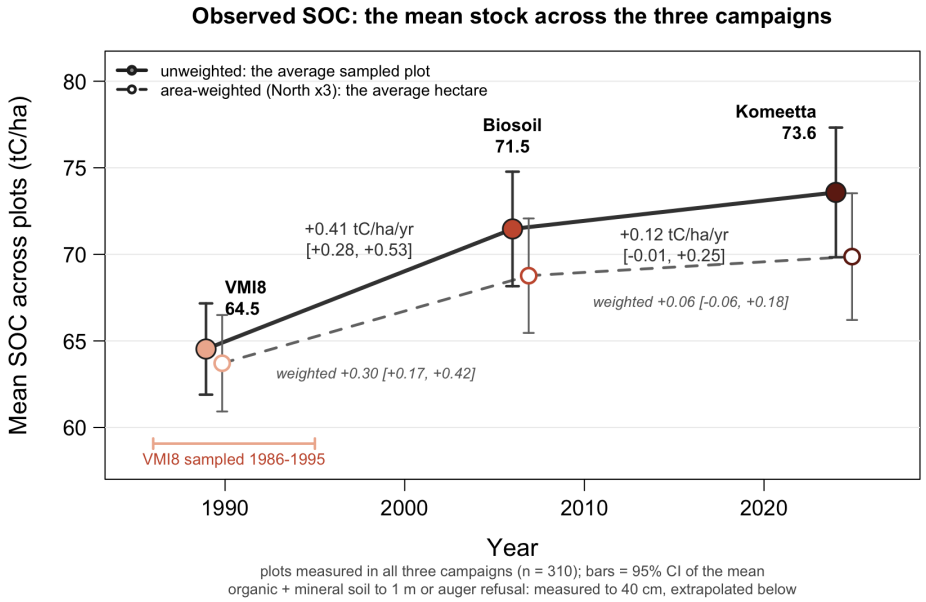


Figure 2: F3_mean_SOCalt — **not in your list: keep?**

1.0.0.6 6. Objectives — the test, and its consequences. Two revisions of the starting state: estimated rather than assumed, and dated back to 1917, as far as the forest record allows. Each is tested against the same models started in balance at the beginning of the measurements and recalibrated on the same data. **And the applied aim, declared as such:** once the assumption is relaxed, what changes in the forecast, and what does that tell us about the future of the Finnish greenhouse gas balance — whether the forest soil can stay a sink, for how long, how it responds to warming, and how much more litter input it would take to keep the sink of the past. Questions:

(i) Can a model started in balance reconcile the soil series, the kinetics and the inputs, and how much of the modelled sink does the history carry when the start is estimated?

(ii) Where does reconciling them place the models on the trade-off between litter input and transit time?

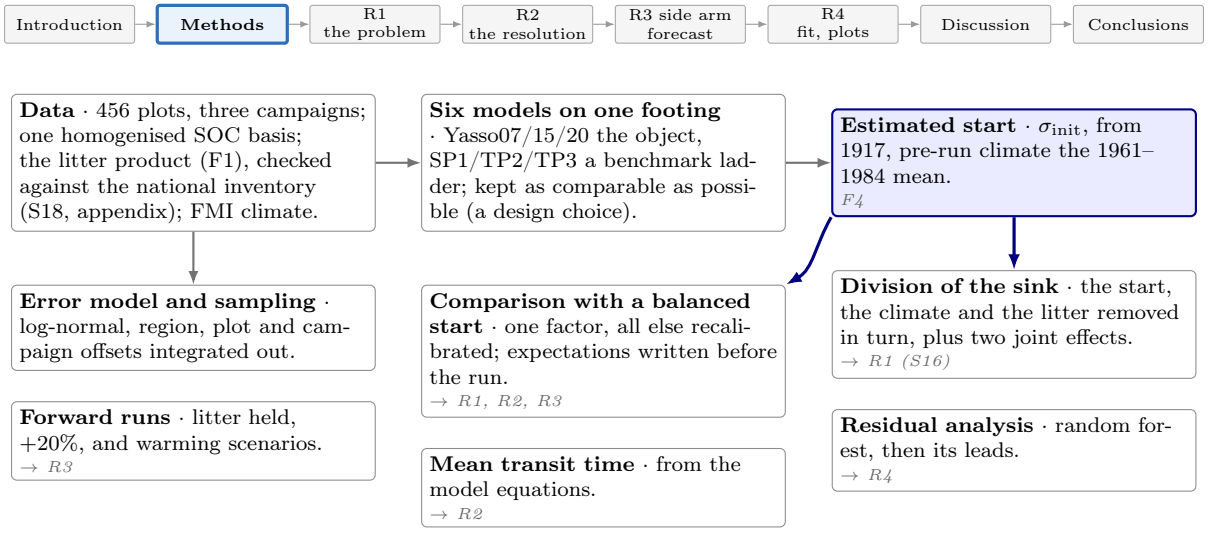
(iii) Once the assumption is relaxed, what follows for the future of the soil sink and of the Finnish greenhouse gas balance: how much room the soil has left, how the sink responds to warming, and how much additional litter input would hold it?

(iv) What do the differences between plots say about the history each plot carries?

Scope of the applied claim, stated where it is declared. The projections describe the calibrated models on the study plots, per hectare: the national inventory's term is not re-estimated, and the baseline projection holds the litter at 2024 and recycles the recent climate. The applied answer is therefore a statement about direction and sensitivity — room left, response to warming, input needed — not a forecast of the reported numbers. National units (Mt CO_2 yr^{-1}) need a sourced area and Aleksi's review.

For the discussion. (a) Paragraph 3 makes the 2021 source an illustration of the general problem — the strongest motivation, and the most delicate for the coauthors; keep, soften, or move after the trilemma? (b) Naming the balanced-start comparison already in the objectives (as above), or only in the Methods?

2 Materials and methods



2.0.0.1 Data. As written. The litter input, one corner of the trilemma, is introduced with F1; where it is introduced, a short paragraph compares it with the national inventory's estimate (already in the draft): on the shared scope (tree turnover and harvest residues) the two agree in the south, where most plots lie, and neither rises after about 2006; the north differs, but holds only 50 of the 456 plots; the inventory also carries natural mortality, which grows and which the product lacks. S18 in the appendix. ▶ F1_inputs

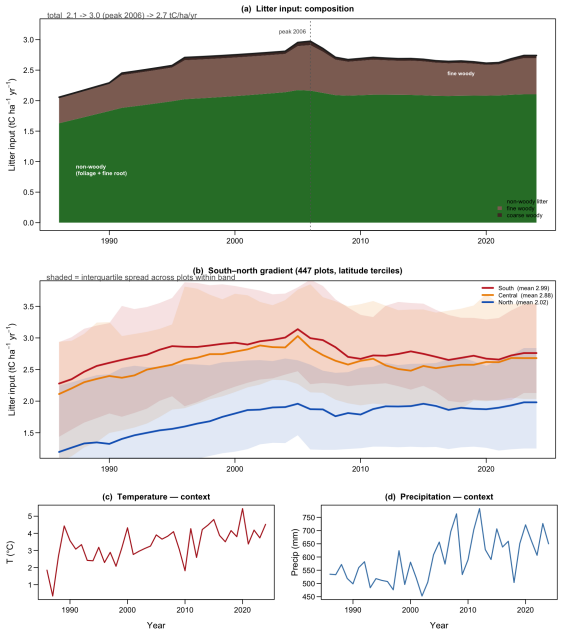
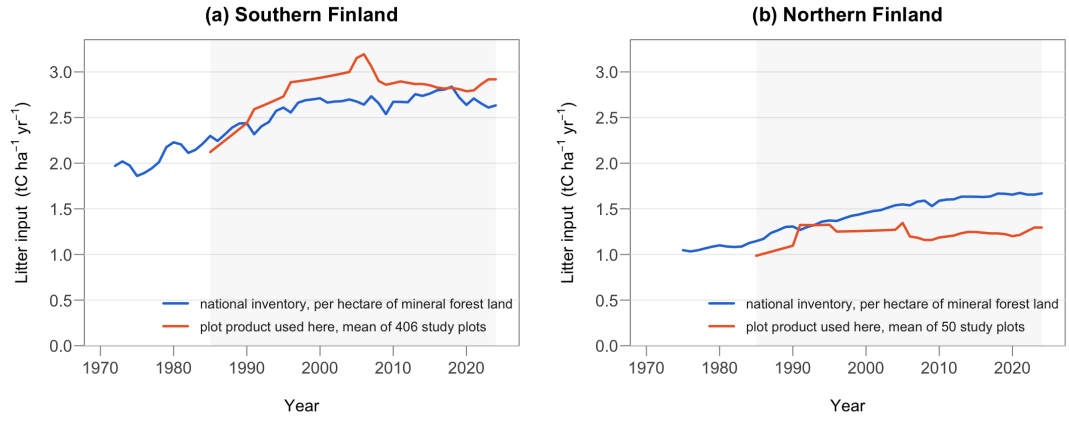


Figure 3: F1_inputs — main text

▶ S18_litter_vs_inventory



Both series: litter from the turnover of living trees and from harvest residues. The inventory's natural-mortality and ground-vegetation terms are excluded. Shaded: 1985-2024.

Figure 4: S18_litter_vs_inventory — **appendix**

▶ F2_baseline

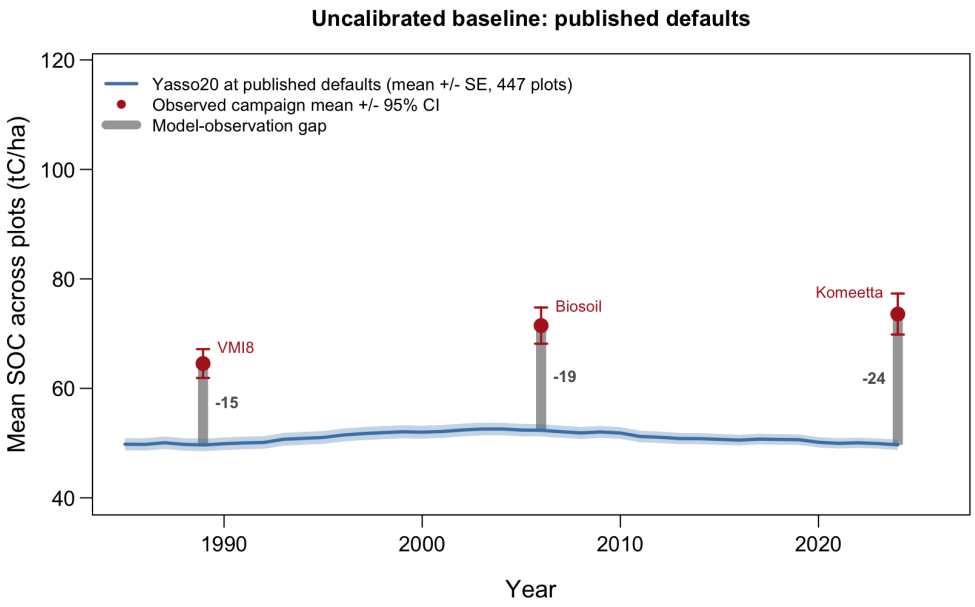


Figure 5: F2_baseline — main text

2.0.0.2 The estimated start. As written, with the pre-run climate now the 1961–1984 mean, so that the soil enters 1985 adjusted to the climate it had lived in. ▶ F4_initialization

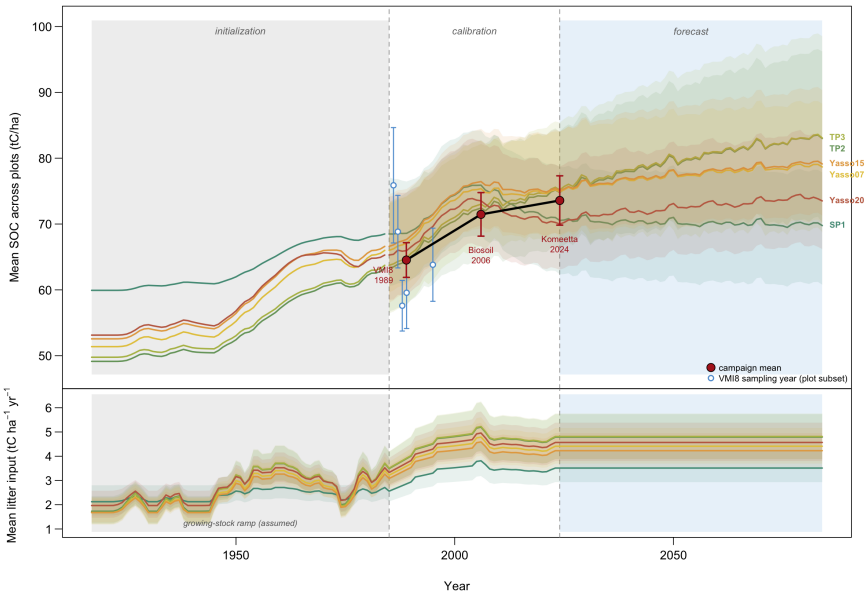


Figure 6: F4_initialization — main text

2.0.0.3 The comparison with a balanced start. The same six models, data, likelihood and priors, with the start fixed at balance in 1985 (the 1985–1989 litter and the pre-run climate) and σ_{init} removed; everything else recalibrated. One factor. Stated as *the steady-state assumption applied at the start of the measurements*, the strongest fair version because the models are free to compensate, and explicitly not as the inventory's procedure. The expectations were written down before the run.

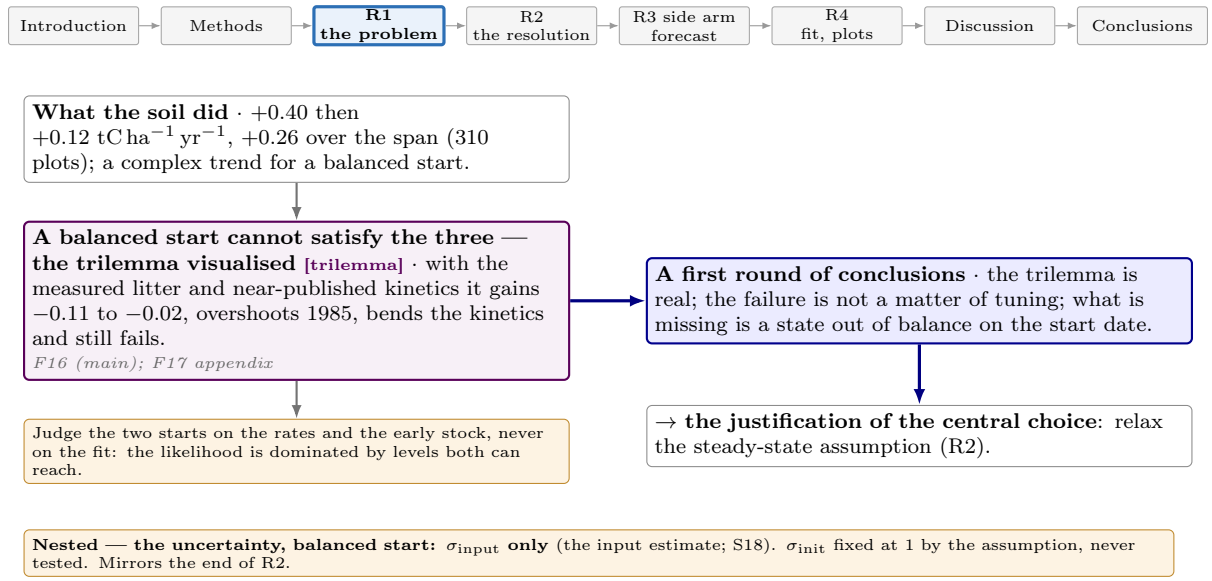
2.0.0.4 The division of the sink. Three runs and two joint effects of climate, in the words of the appendix.

> **Say in the Methods that the expectations were written before the run, and that two failed** (the balanced start overshoots the 1985 stock; its litter multiplier went down, not up)?

3 Results and discussion

R1 — THE OPENING: THE PROBLEM, TESTED THE HARD WAY

> **Role (Lorenzo, 2026-10-12).** R1 is the opening, a better definition of the problem. We know the conceptual problem of a balanced start, and we know the data hold a complex trend that could trouble it; here we test it the hard way and reach a first round of conclusions. From R1 to R2 runs the justification of the central choice.



3.0.1 What the soil did

3.0.1.1 +0.40 tC ha^{-1} yr^{-1} over the first interval, +0.12 over the second, +0.26 over the span (310 plots, unweighted, true sampling years). The second interval is the fragile one (its interval includes zero); the span is robust.

3.0.2 A balanced start cannot satisfy the three — the trilemma visualised

3.0.2.1 The first result, and the moment the trilemma of the Introduction becomes visible. Started in balance in 1985 and recalibrated, all six models lose carbon over 1985–2024, −0.11 to −0.02, against the observed +0.26. They rise with the litter to 2006 and give the rise back after it, and they start about 9 tC ha^{-1} above the first campaign (73–76 against 64.5), because a soil in balance with the 1985 litter holds more carbon than the soil held. Over 1985–2006 none reaches the observed rise (+0.06 to +0.20 against +0.40); over 2006–2024 all six lose carbon. F16 shows it against the campaigns: the two starts differ mainly at the first campaign, and separate again in the projection. ▶ F16_eqinit_trajectories

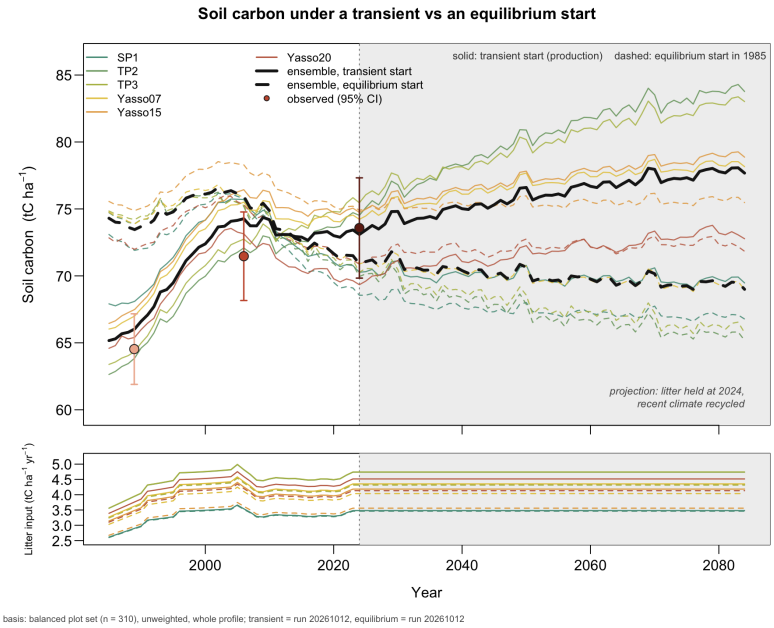


Figure 7: F16_eqinit_trajectories — main text

▶ F17_eqinit_rates

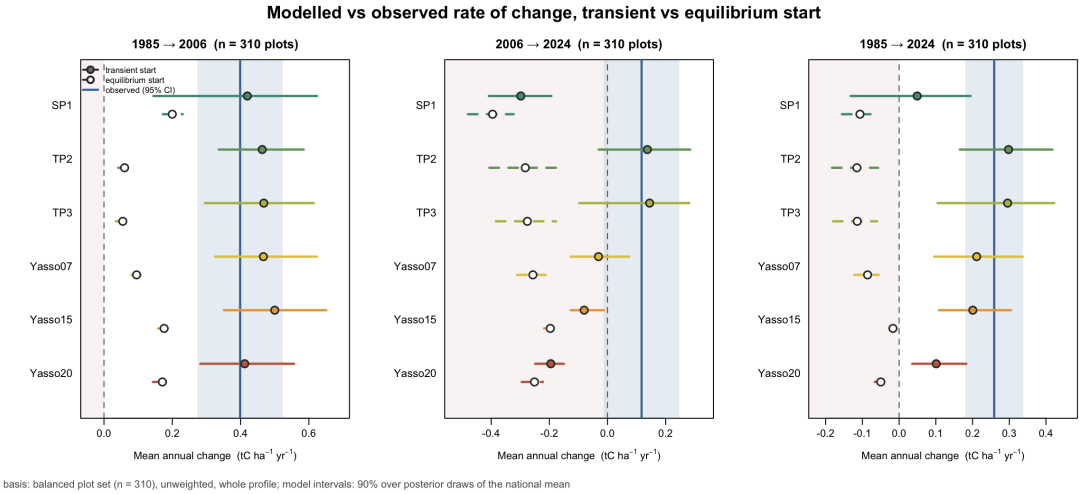


Figure 8: F17_eqinit_rates — **appendix**

3.0.2.2 ⇝ **The trilemma, contextualised.** The text names it here: held to the measured litter and to kinetics near the published ones, the balanced start gives up the soil series — the corner of the triangle the Introduction predicted. It is free to bend the kinetics and does — less input, faster and more temperature-sensitive decomposition (transit time 22–27 → 17–24 yr) — and still cannot produce the accumulation. How far the kinetics would have to move is the corner the comparison leaves partly open.

Do not judge the two starts by the fit. The likelihood is dominated by the stock levels, which both starts reach (log-likelihoods within 8, the estimated start better in five of six); the rates and the early stock discriminate. Say so in two sentences before a reader concludes the starts are equivalent.

Close the third corner? A balanced start with the rate priors widened would show how far the kinetics must move to fit both series; one factor, six jobs. Or state it as a limitation.

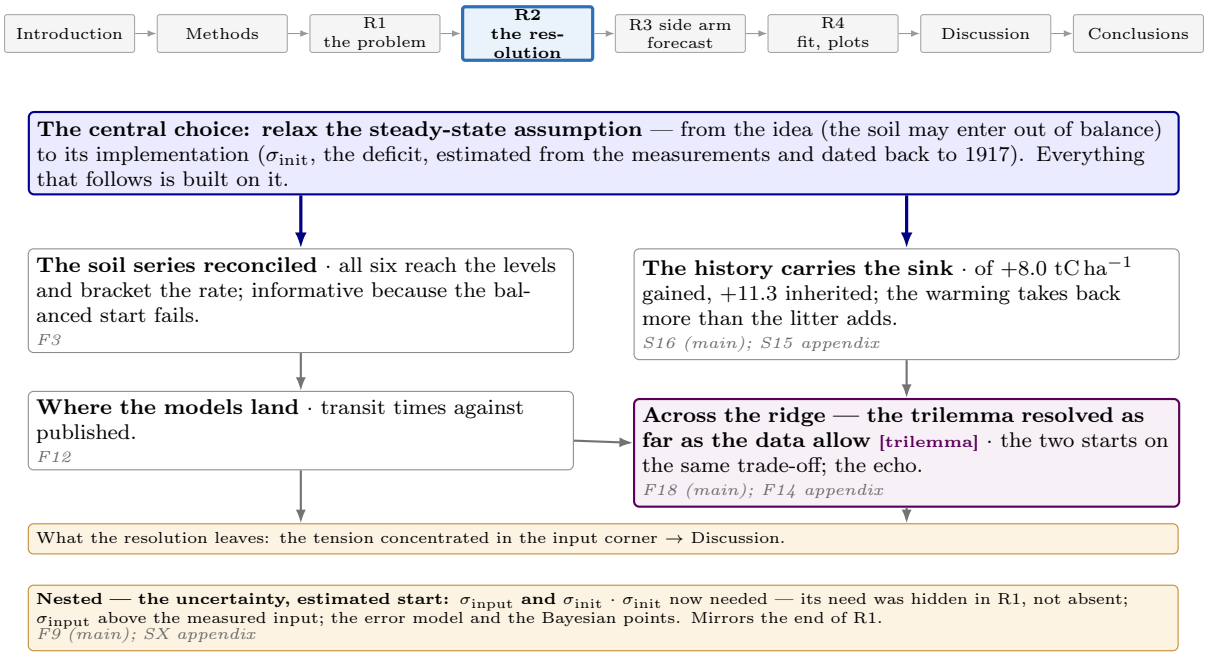
3.0.3 A first round of conclusions — and the justification of the central choice

3.0.3.1 Three things are settled before anything else is built. (1) The trilemma is real on these data: with the measured litter and near-published kinetics, a soil assumed balanced in 1985 cannot reproduce the measured accumulation. (2) The failure survives the freedom to compensate, so it is not a matter of tuning. (3) What the balanced start lacks is a state that is not in balance on the start date. **That is the justification for the central choice of the paper**: to relax the steady-state assumption and estimate the starting state from the measurements, which is what R2 does.

3.0.4 Nested: the uncertainty, as the balanced start sees it — σ_{input} only

3.0.4.1 The same side arm as at the end of R2, with one parameter. The balanced start carries one auxiliary uncertainty parameter, σ_{input}, which absorbs the uncertainty of the input estimate — the point where the data discussion (the litter product against the national inventory, S18) enters the calibration. In the balanced start it relaxes (Yasso15 1.54 → 1.31, TP2 1.75 → 1.60), while the kinetics strain. σ_{init} **is not absent: it is fixed at** 1 **by the steady-state assumption** — a value never questioned and never tested, only assumed. Implementation details here: the same error model and priors as R2, the fit equivalence of the two starts (the likelihood dominated by levels) and why the comparison is made on the rates.

R2 — THE RESOLUTION: RELAXING THE STEADY-STATE ASSUMPTION



Role (Lorenzo, 2026-10-12). The main arc, carrying the most load. Relaxing the steady-state assumption is the paper's own approach to the trilemma — the proposed solution, and *the central choice*. It moves the models across the ridge in parameter space, and the trade-off translates according to the new assumption. Everything that follows (the forecast, the plot analyses) is built on it, once it is justified; its ramifications are examined, but the choice is the centre.

3.0.5 The central choice

3.0.5.1 From the general idea to its implementation. The idea: the soil may enter the observations out of balance. The implementation: the start is estimated from the measurements and dated back to 1917, through one quantity, σ_{init}, the deficit the soil carries — the parameter the balanced start had fixed at 1. The same data, priors and likelihood. What σ_{init} means, and what it costs, is in the nested part at the end of R2.

3.0.6 The estimated start reconciles the soil series

3.0.6.1 All six reach the three campaign levels. Over 1985–2024 they span +0.05 to +0.30 against the observed +0.26, four of six inside its interval (SP1 +0.05, Yasso20 +0.10 below). Over 1985–2006 all six reach the observed rise; over 2006–2024 only TP2 and TP3 reach the observed +0.12, Yasso07 is near zero (−0.03), and SP1, Yasso15 and Yasso20 lose carbon. ▶ F3_mean_soc

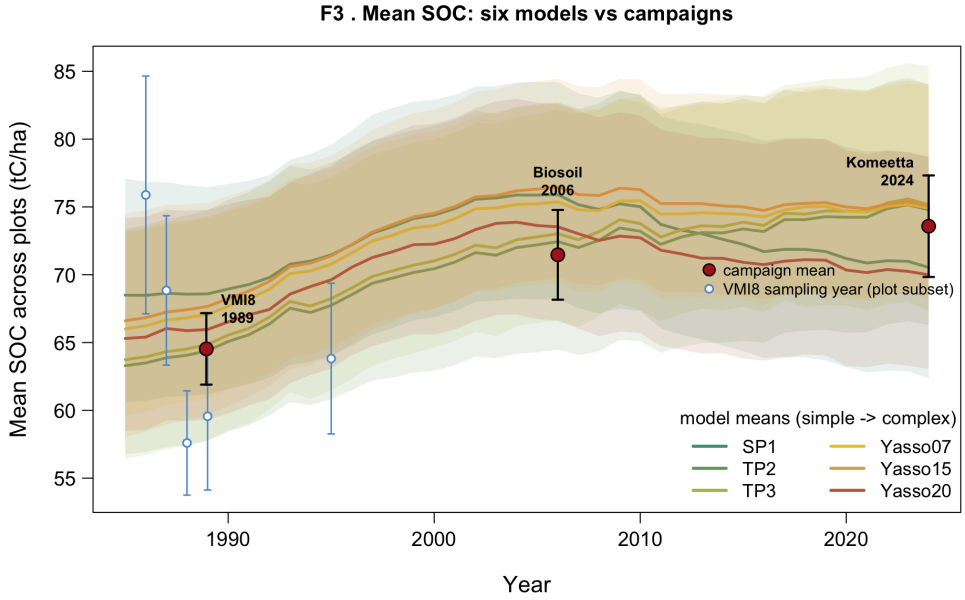


Figure 9: F3_mean_soc — main text

3.0.6.2 ⇝ **Bracketing is informative because it is not automatic.** Matching the rate would mean little if any calibration on the same data could do it. The balanced start, calibrated on the same measurements with one degree of freedom fewer, fails in every model.

3.0.7 And the modelled sink is the history

3.0.7.1 The division of the sink. Six-model average, 1985–2024: of +8.0 tC ha^{-1} gained, the deficit inherited from before 1985 carries +11.3; the litter change since 1985 adds +8.9; the warming

since the 1961–1984 reference removes −12.2. The inherited part exceeds the whole sink in all six models. ▶ S16_attribution_ensemble_smooth10

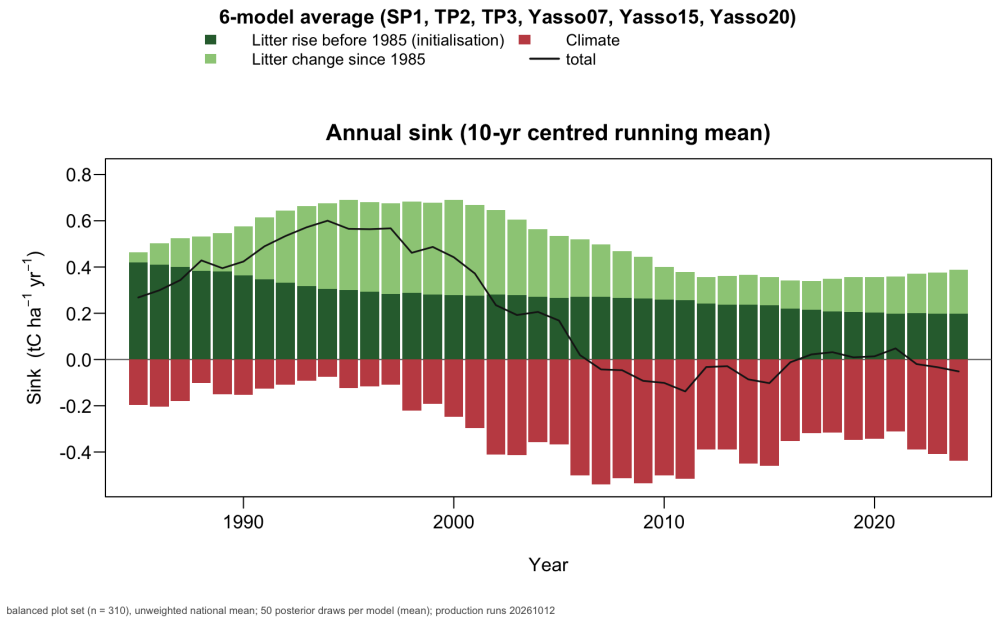


Figure 10: S16_attribution_ensemble_smooth10 — main text

▶ S15_attribution_by_model_smooth10

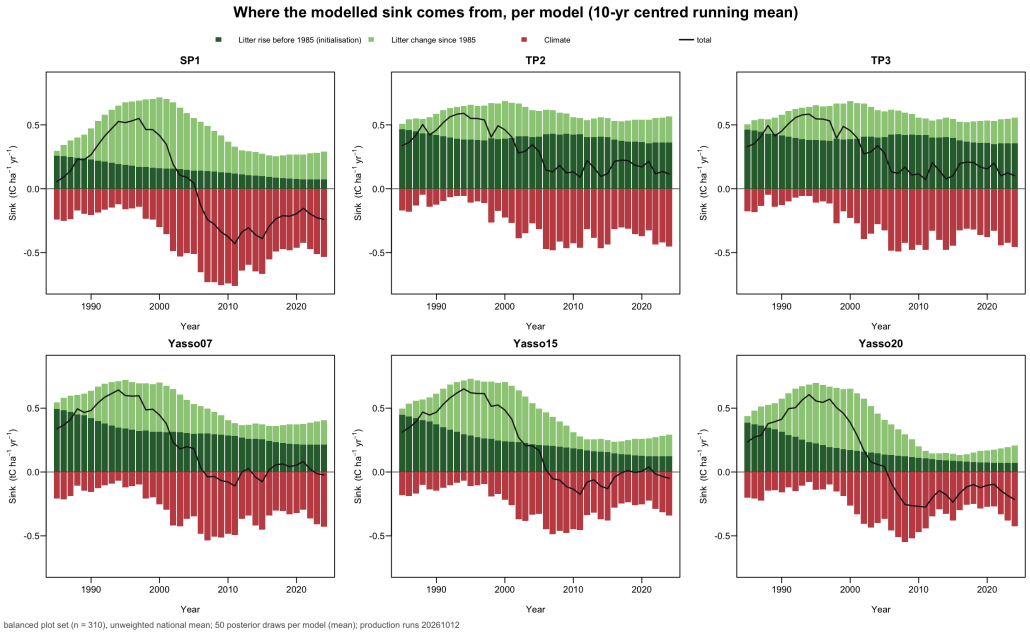


Figure 11: S15_attribution_by_model_smooth10 — **appendix**

3.0.7.2 ⇝ **Necessary, and dominant.** The balanced start shows that the history is *necessary* — with everything else free, the models cannot do without it. The division shows how much it *weighs* — more than the whole sink, because the warming takes back more than the litter rise adds. The

second answers the objection to the first ("the inherited part is large only because the calibration chose a low σ_{init}"): the data demand the deficit.

The numbers moved with the 1961–1984 pre-run climate (2026-10-13), but not as expected: the inherited part *grew* (+9.3 → +11.3), the climate part grew more (−7.5 → −12.2, all warming since the colder reference now counts), and the total fell (+10.5 → +8.0). The robust statement is the ordering — the sink inherited; litter and warming of the same order and opposite sign. The size of the climate part depends on the reference climate; say so.

S16 in three parts or five terms (the two joint effects of climate as hatched bands)?

3.0.8 Where the models land: shorter transit times than published

3.0.8.1 Calibrated against published: Yasso07 and Yasso15 turn over faster, Yasso20 sits on its published value. ▶ F12_mrt_yasso

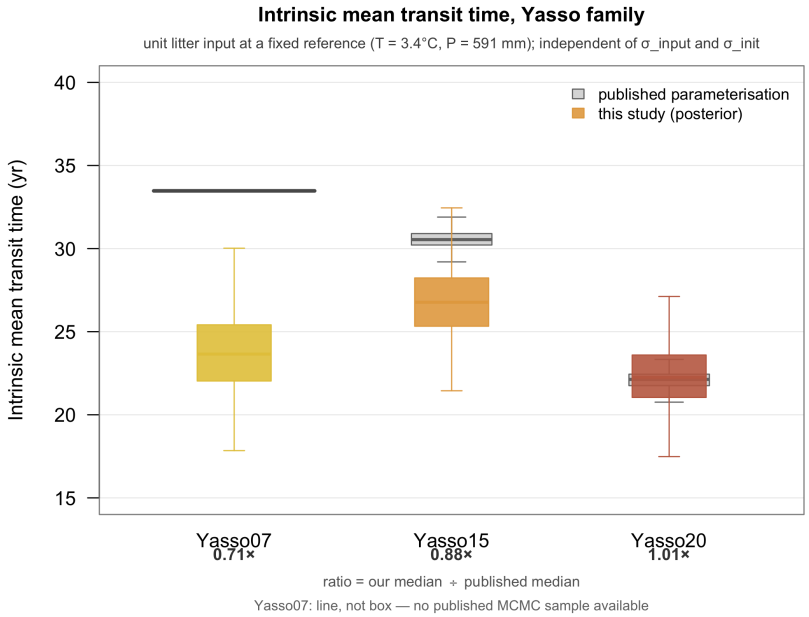


Figure 12: F12_mrt_yasso — main text

3.0.9 Across the ridge — the trilemma resolved as far as the data allow

3.0.9.1 The soil fixes the product of input and transit time, and the posteriors ride that trade-off. The two starts land at different places on it: in five of the six models the balanced start sits at less input and faster turnover (SP1, a single slow pool, barely moves). ▶ F18_eqinit_tradeoff

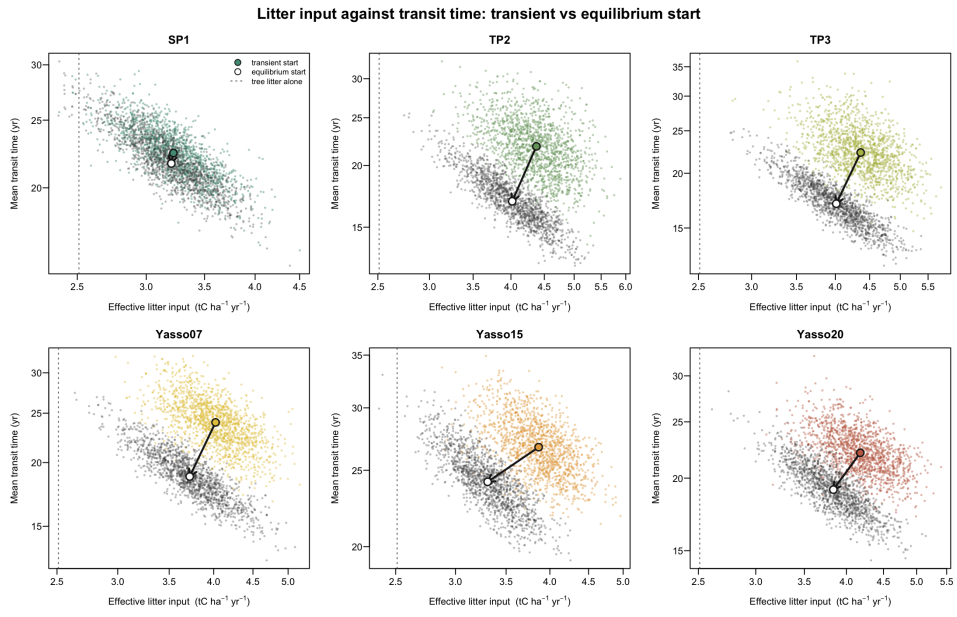


Figure 13: F18_eqinit_tradeoff — main text

▶ F14_mrt_ridge

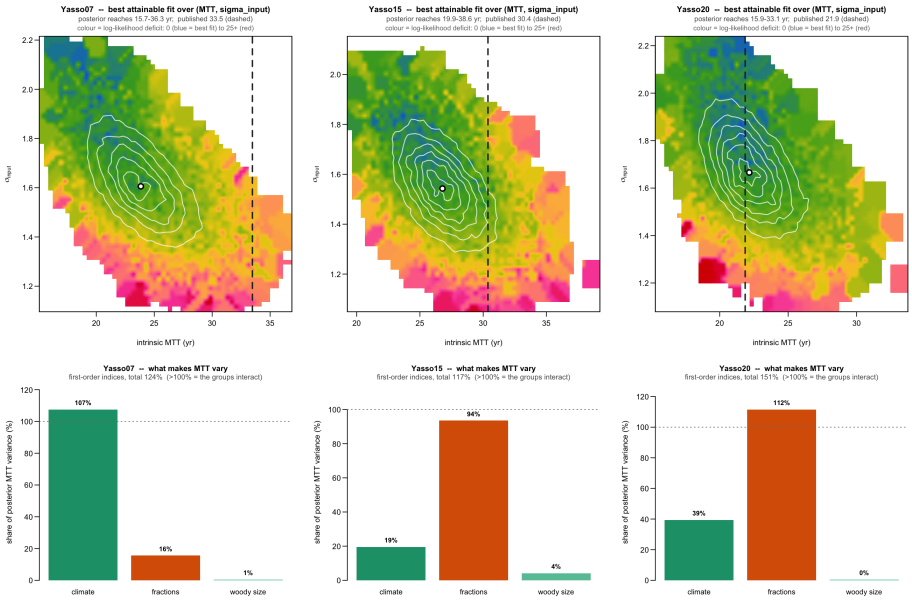


Figure 14: F14_mrt_ridge — **appendix**

3.0.9.2 ⇝ **The echo, in parameter space. The triangle comes back once more**, after F16 showed it in the trajectories. The soil series is the ridge; the measured input (the dotted line) and the published kinetics sit off it; a balanced start can stand on at most two of the three, and the estimated start, carrying the history, stands somewhere else on the same ridge. Relaxing the assumption moves the models across it, and the trade-off translates according to the new assumption. The text should say it in one or two sentences — after all the work, the trilemma is visible — without redrawing the

figure.

3.0.9.3 ⇝ **What the resolution leaves.** The tension is not removed but concentrated in one corner: the soil series is met and the kinetics stay near the published values (Yasso20 on its own, Yasso07 and Yasso15 faster), while the input the models need stays above what the product supplies. That remaining gap is the thread to the Discussion of the inputs the product may miss.

3.0.10 Nested: the uncertainty, as the estimated start sees it — σ_{input} and σ_{init}

3.0.10.1 The same side arm as at the end of R1, now with two parameters. σ_{init} becomes needed, and **that need was there before**: in R1 it was hidden by forcing the steady-state assumption, which fixes it at 1. R2 may look more uncertain, but it only makes explicit a quantity R1 assumed without testing. The two auxiliary parameters share a width by design.

- σ_{input} — more input than measured: effective flux 3.22–4.40 against 2.511, inside the envelope of boreal production, set by the likelihood rather than the prior (the posterior stays above the prior centre whichever anchor is used). The national inventory's full estimate is 1.08–1.26× the product on the southern plots, bracketing the understorey-corrected anchor (1.21) and below the posteriors (1.28–1.75); SX sets it against independent estimates.
- σ_{init} — the deficit the data demand: 0.45–0.56 in five of six models (SP1 0.78), lower than the growing-stock record implies (~ 0.90), so read as an *effective starting deficit of the soil* (slash-and-burn, raking, heavy cutting) rather than a ratio of litter inputs. It carries the forecast: headroom correlates with it at -0.76 to -0.97 (R3).
- **The implementation, and its advantages**: the error model (region, plot and campaign offsets integrated out, a declared total of 0.800), what it buys (an honest number of independent observations, 221), and the Bayesian points — priors as information, homogenised across models; the posterior against the prior.

▶ F9_auxiliary_sigmas

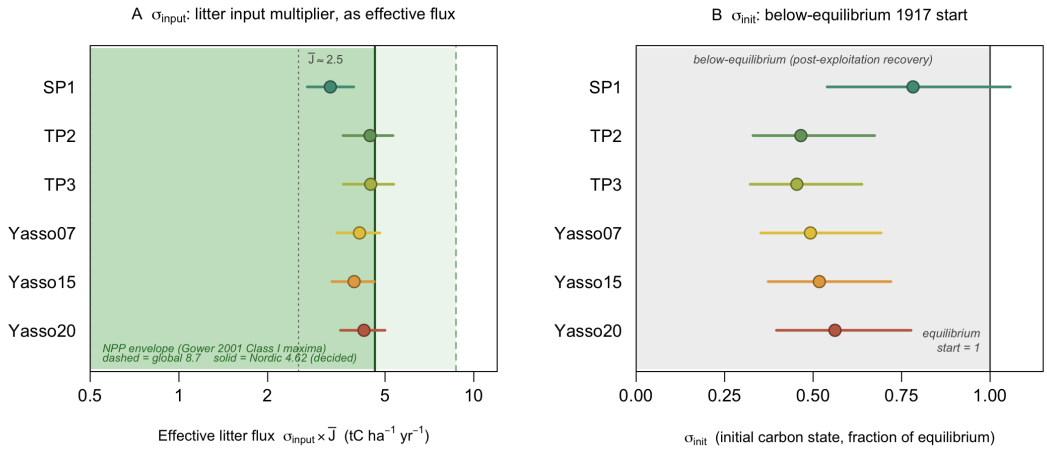


Figure 15: F9_auxiliary_sigmas — main text

▶ SX_input_vs_literature

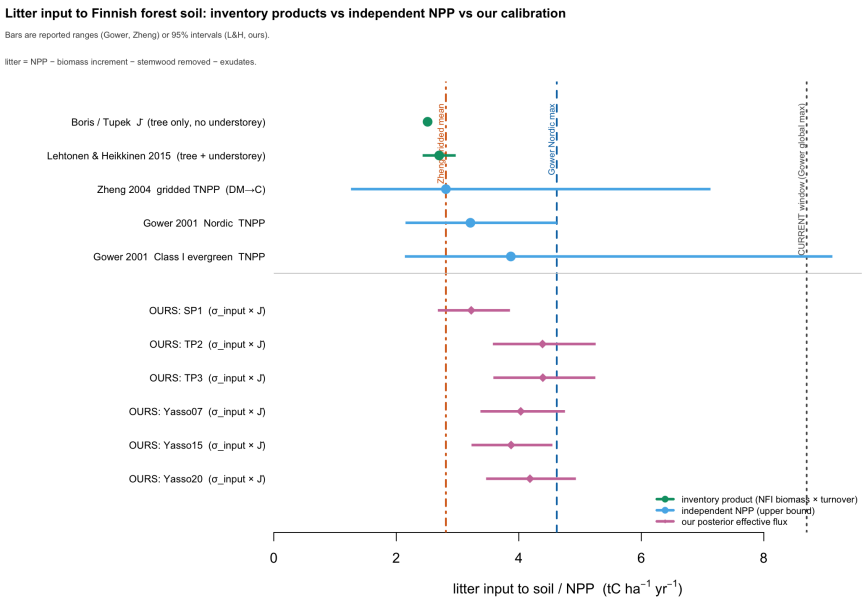


Figure 16: SX_input_vs_literature — **appendix**

The direction depends on the comparator, so the title names none. Against the published parameters the calibration has faster turnover (Yasso07, Yasso15); against the balanced start, slower. "Displaces along" covers both; never add a direction.

R3 — SIDE ARM: WHAT IT MEANS FOR THE FORECAST AND THE GHG BALANCE

3.0.10.2 The applied answer to objective (iii). Built on the central choice of R2: what the relaxed assumption says about the future of the soil sink — room left, the input that would hold it, the response to warming — and so about the Finnish greenhouse gas balance, within the scope declared in the objectives.

Introduction → Methods → R1 the problem → R2 the resolution → **R3 side arm forecast** → R4 fit, plots → Discussion → Conclusions

Headroom only with the history · balanced start: sink ≈ 0; estimated: +0.07–0.18, headroom +21–40%.
prose; F19 appendix

→

How much input would hold the sink · about +10% for 20 years (Yasso), half for 10; the recent rate needs almost nothing.
table

→

Warming · −5 to −7 tC ha^{-1} by 2084 (SSP2-4.5); only Yasso07 depends on where it lands.
F15 (main)

The held-litter projection contains no climate change; the input numbers are lower bounds under warming.

3.0.11 Headroom only with the history

3.0.11.1 In prose; the figure is in the appendix. Projected over 2025–2044 with the litter held at 2024 and the recent climate recycled, the balanced start gives a sink of −0.12 to +0.02 tC ha^{-1} yr^{-1} and a headroom of −8 to +9%; the estimated start gives +0.07 to +0.18 and +21 to +40% (SP1 ≈ 0) — the disequilibrium Liski et al. (2006) found by another route (+38%). The general claim of the Introduction, measured: the starting state decides whether the soil has room left. ▶ F19_eqinit_forecast

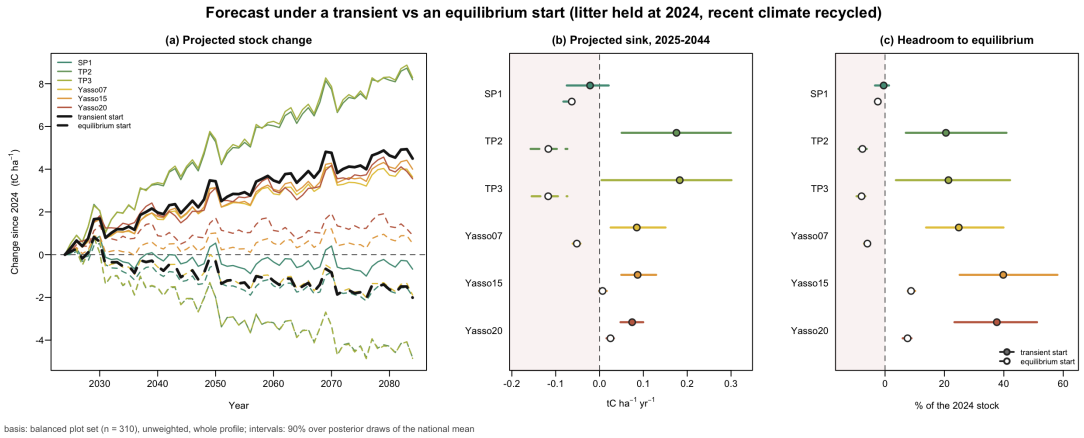


Figure 17: F19_eqinit_forecast — **appendix**

3.0.12 How much input would hold the sink?

3.0.12.1 The question a ministry would ask. The models are linear in their input, so the projections with the litter held and with +20% answer it exactly for any target sink (`doublechecks/input_to_hold_sink.R`

target sink	window	Yasso07	Yasso15	Yasso20	TP2 / TP3	SP1
observed 1985–2024 (+0.26)	10 yr	+8%	+7%	+5%	+5 / +5%	+11%
	20 yr	**+11%**	**+10%**	**+11%**	+7 / +6%	+13%
each model's own 1985–2024	20 yr	+9%	+7%	+3%	+10 / +10%	+4%
observed 2006–2024 (+0.12)	20 yr	+2%	+2%	+3%	−5 / −5%	+6%

3.0.12.2 ⇝ About a tenth more input, sustained, holds the long-term sink for twenty years (0.4–0.5 tC ha^{-1} yr^{-1} of effective input in Yasso; each +10% buys 0.15–0.24 tC ha^{-1} yr^{-1} of sink). The gain lasts only while the input does, and it postpones saturation rather than removing it.

Lower bounds, and provisional: no further warming is in these projections. The required input depends on both the starting state (the baseline sink) and the transit time (the gain per unit of input) — the paper's two main results meet here.

Which target, the observed long-term rate, the recent one, or each model's own? A warming-corrected column for Yasso? National units (Mt CO_2 yr^{-1}) need a sourced area and Aleksi's eye.

3.0.13 Warming, and whether it depends on where the calibration lands

3.0.13.1 Under warming to SSP2-4.5 the stock falls by 5–7 tC ha^{-1} by 2084 in all three Yasso versions; only Yasso07 separates the published and calibrated parameters (−2.22 tC ha^{-1}), because its single climate modifier ties turnover to temperature sensitivity. For the inventory, which runs Yasso07, that is the applied point. ▶ F15_forward_arms

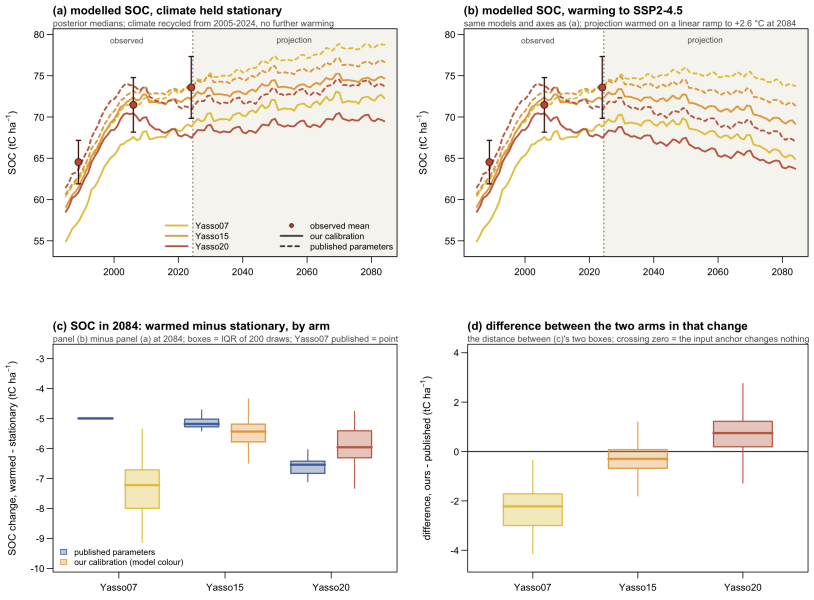
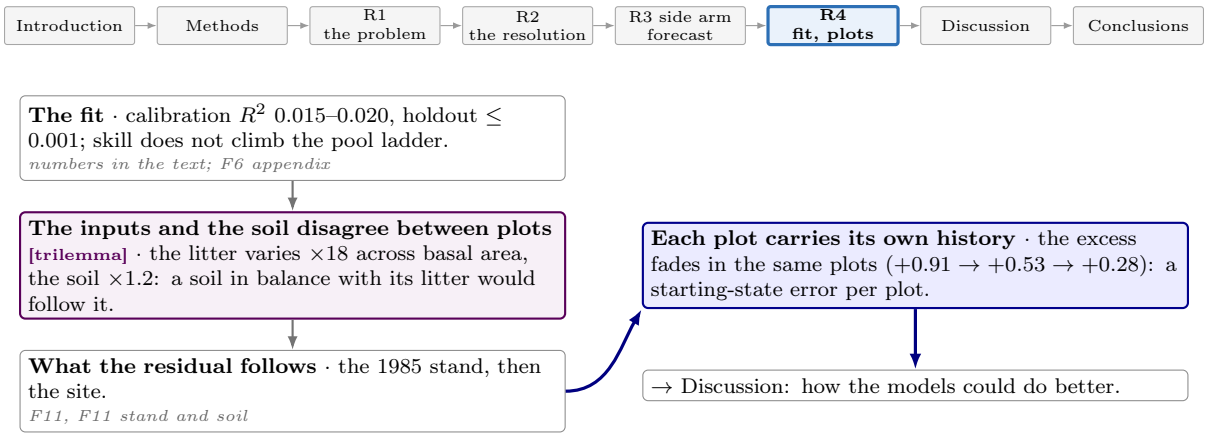


Figure 18: F15_forward_arms — main text

R4 — THE FIT, AND THE PLOTS



3.0.14 The fit

3.0.14.1 Calibration R^2 0.015–0.020, holdout $R^2 \leq 0.001$; skill does not climb the pool ladder. The models reproduce the national mean and almost none of the differences between plots. Numbers in the text; F6 in the appendix. ▶ F6_obs_vs_pred_holdout

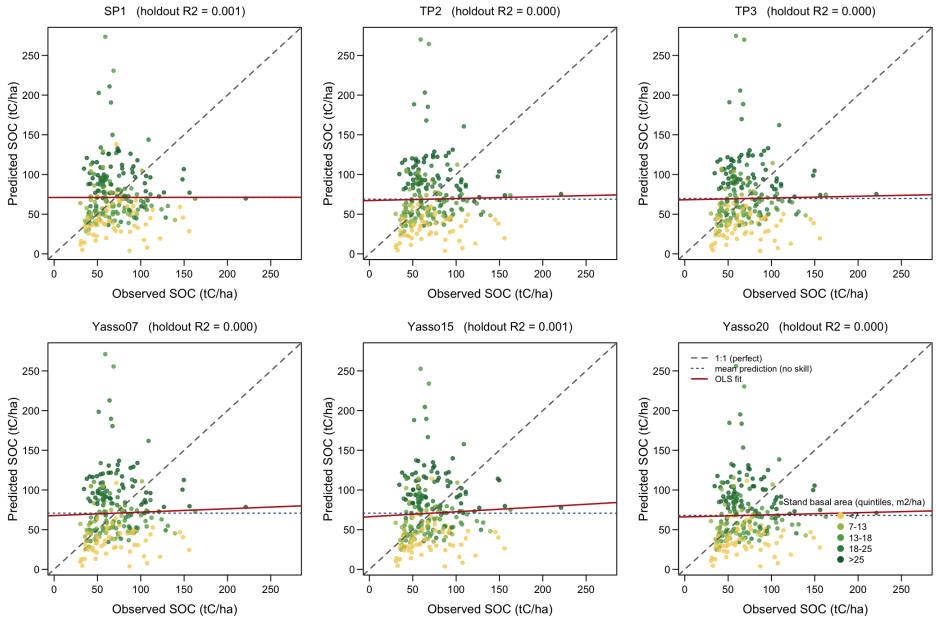


Figure 19: F6_obs_vs_pred_holdout — **appendix**

3.0.15 The inputs and the soil disagree between plots

3.0.15.1 The trilemma at plot scale. The litter product varies about ×18 across the basal-area range; the soil about ×1.2. A soil in balance with its litter would follow it, so the gap is the inputs corner of the trilemma seen plot by plot — one sentence pointing back, not a second echo.

3.0.16 What the residual follows: each plot carries its own history

3.0.16.1 The residual follows the 1985 stand, then the site (F11 → F11 stand and soil). In the same plots it fades over the three campaigns (+0.91 → +0.53 → +0.28 for sparse 1985 stands): the signature of a wrong starting state, not of a missing input. One national pre-run shape cannot carry each stand's history. ▶ F11_rf_heatmap

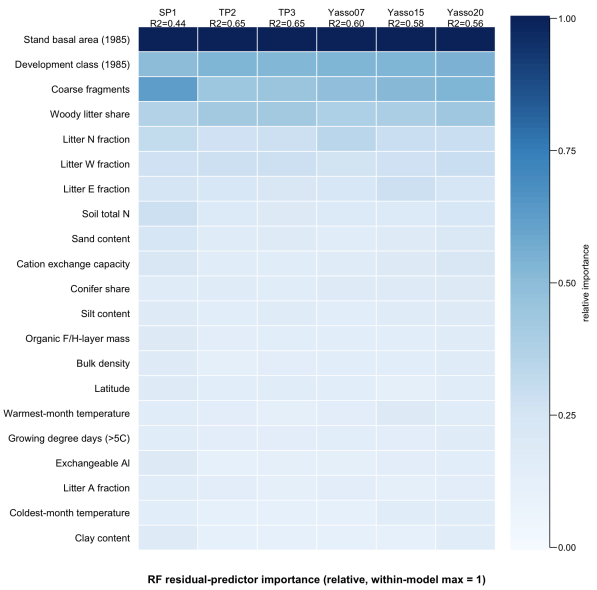


Figure 20: F11_rf_heatmap — main text

▶ F11_stand_and_soil

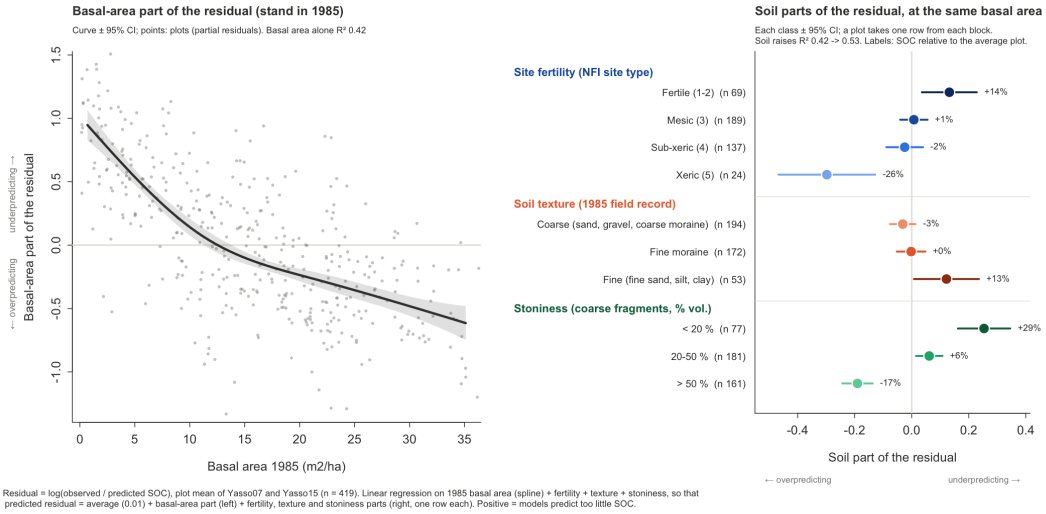


Figure 21: F11_stand_and_soil — main text

4 Discussion — how the models could do better

4.0.0.1 1. A starting state per plot. The plot-level error fades at about the models' turnover pace: a start built from each plot's own history (the last regeneration cut dated from the 1985

stand age) is the natural next model step. The excess lives in stands young in 1985 (F11b). ▶ F11b_time_since_cut

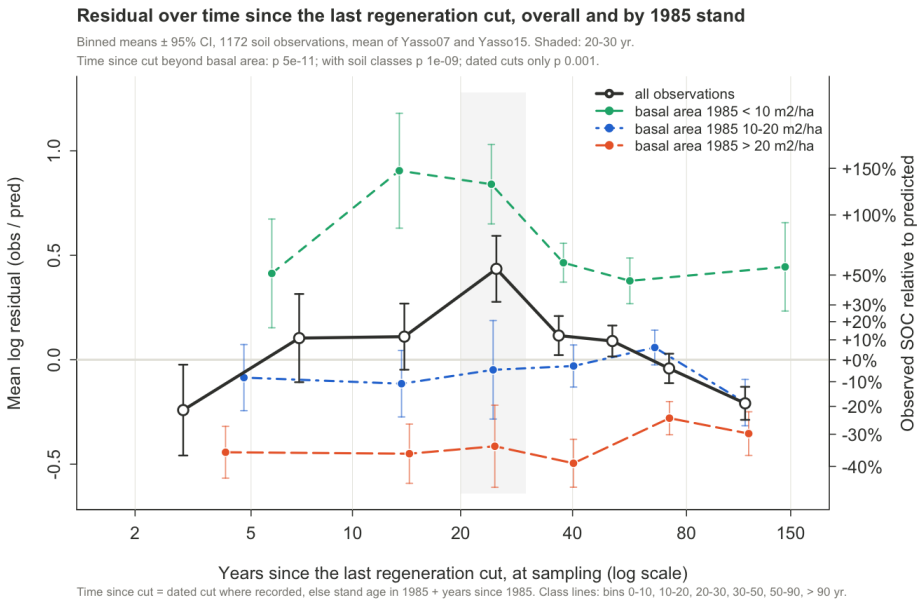


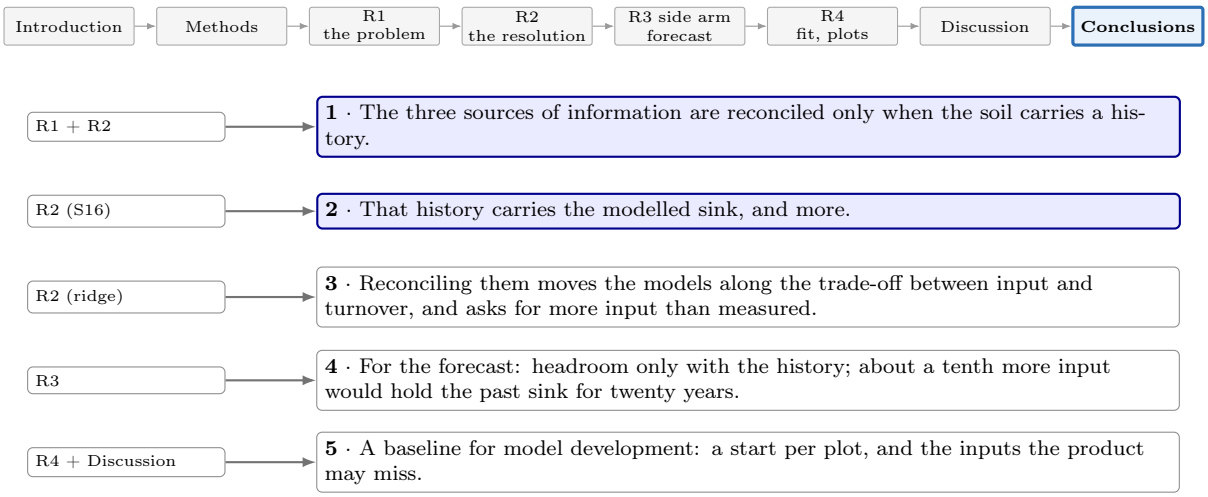
Figure 22: F11b_time_since_cut — **appendix**

4.0.0.2 2. The inputs the litter product may miss. Natural mortality — now quantified in the national inventory, 0.1–0.16 tC ha^{-1} yr^{-1} by now in the south; roots and rhizodeposition; understorey; mycorrhizal fungi; fine-root lifespan (pointing the other way); early-succession vegetation. A set that brackets the gap, never a sum that matches it.

4.0.0.3 3. Pioneer vegetation, where the two meet. The excess carbon of the stands young in 1985 was built during early stand development, either inherited or supplied by vegetation no tree-based product records; the data do not separate the two. Lauri Mehtätalo (LUKE) to be cited **[OPEN:** which work**]**.

4.0.0.4 4. The site, briefly. Fertility, texture and stones explain a further tenth of the residual **[OPEN:** check whether the stone effect is partly the stone correction of the observed stocks**]**.

5 Conclusions



5.0.0.1 **1.** The soil series, the kinetics and the inputs are reconciled only when the soil carries a history: a balanced start cannot, an estimated start brackets the measured accumulation.

5.0.0.2 **2.** That history carries the modelled sink of 1985–2024, and more: the warming takes back more than the litter rise adds.

5.0.0.3 **3.** Reconciling the three moves the models along the trade-off between input and turnover, and asks for more input than the litter product supplies.

5.0.0.4 **4.** For the forecast and the Finnish greenhouse gas balance: the soil keeps room to gain only with the history, finite and costly to realise; about a tenth more litter input would hold the past sink for twenty years; warming takes part of it back, and for the inventory's own model the size of that response depends on where the calibration lands.

5.0.0.5 **5.** A baseline for further model development: a start per plot, and the inputs the product may miss. (The national-inventory implications go just before the Limitations, not here.)

5.0.0.6 **Bonus product.** The maps.

Open questions collected

1. The weight budget (30 / 25 / 20 / 10 + Discussion 15).
2. F10b and F3_mean_SOCalt in the Introduction.
3. The 2021 source as the illustration in Introduction 3: keep, soften or move.
4. The balanced-start comparison named in the objectives, or only in the Methods.
5. The written expectations and the two that failed, in the Methods.
6. Close the third corner of the trilemma with a run, or state it as a limitation.
7. S16 in three parts or five terms.
8. The target sink in the input table; a warming-corrected column; national units.

9. The size of the climate part of S16 depends on the reference climate (1961–1984): how to state it.